Google Cloud Platform + AI Fundamentals

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Compute Engine

Run large-scale workloads on virtual machines hosted on Google's infrastructure

App Engine

A platform for building scalable web apps and mobile backends

Container Engine

Run Docker containers on Google's infrastructure, powered by Kubernetes

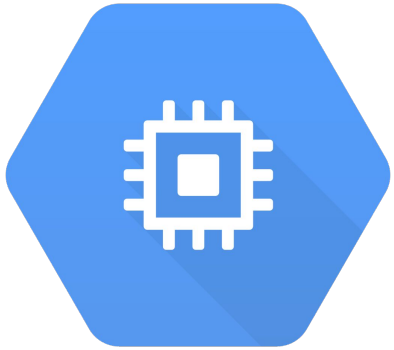
Cloud Run

Fully managed compute platform for deploying and scaling containerized applications quickly and securely.

Cloud Functions

A serverless platform for building event-based microservices triggered by events in GCP

Compute Engine



- La alternative de GCP a EC2 de Amazon o las máquinas virtuales de Azure
- Ofrece máquinas virtuales completamente configurables, con SLAs y precios acordes al mercado
- Se organizan en familias, series y tipos concretos para que sea lo más fácil posible elegir la configuración que se adapte a nuestras necesidades

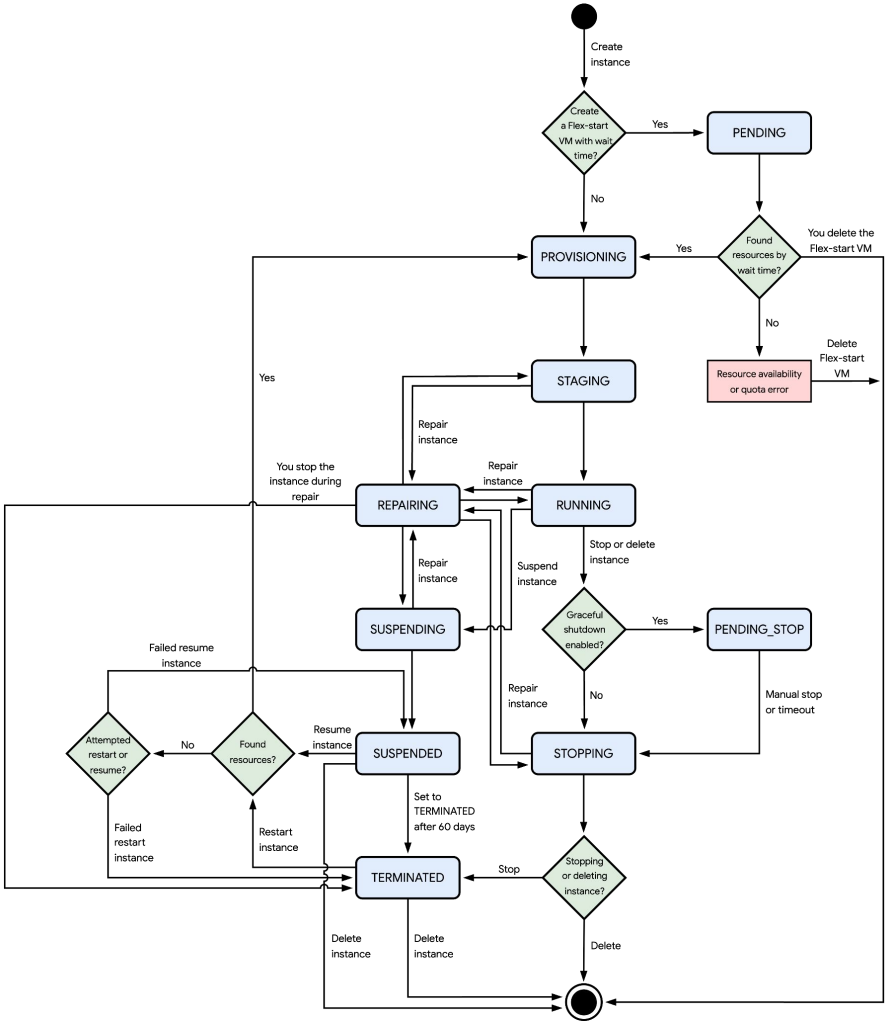
Optimized workloads			
Storage-optimized	**Compute-optimized**	**Memory-optimized**	**Accelerator-optimized**
Z3	**H4D, H3, C2 and C2D**	**X4, M4, M3, M2, M1**	**A4X Max, A4X, A4, A3, A2, G4, G2**
Highest block storage to compute ratios for storage-intensive workloads	Highest performance and lower cost for high performance computing (HPC), multi-node and compute-bound workloads	Highest memory to compute ratios for memory-intensive workloads	Optimized for accelerated high performance computing workloads

• SQL, NoSQL, and vector databases
• Data analytics and data warehouses
• Search
• Media streaming
• Large distributed parallel file systems
	• Manufacturing, weather forecasting, electronic design automation (EDA), High-performance web servers
	• Healthcare and life sciences, scientific computing
	• Seismic processing and structural mechanics applications
	• Modeling and simulation workloads, AI/ML
	• High-performance web servers, Game Servers
		• Small to extra-large SAP HANA in-memory databases
		• In-memory data stores, such as Redis
		• Simulation
		• High Performance databases such as Microsoft SQL Server, MySQL
		• Electronic design automation
			• Generative AI models such as the following:
			  • Large Language Models (LLM)
			  • Diffusion Models
			  • Generative Adversarial Networks (GAN)
			• CUDA-enabled ML training and inference
			• High-performance computing (HPC)
			• Massively parallelized computation
			• BERT natural language processing
			• Deep learning recommendation model (DLRM)
			• Video transcoding
			• Remote visualization workstation

General-purpose workloads			
N4, N4A, N4D, N2, N2D, N1	C4, C4A, C4D, C3, C3D	E2	Tau T2D, Tau T2A
Balanced price/performance across a wide range of machine types	Consistently high performance for a variety of workloads	Day-to-day computing at a lower cost	Best per-core performance/ cost for scale-out workloads
• Medium traffic web and app servers • Containerized microservices • Business intelligence apps • Virtual desktops • CRM applications • Development and test environments • Batch processing • Storage and archive	• High traffic web and app servers • Databases • In-memory caches • Ad servers • Game Servers • Data analytics • Media streaming and transcoding • CPU-based ML training and inference	• Low-traffic web servers • Back office apps • Containerized microservices • Microservices • Virtual desktops • Development and test environments	• Scale-out workloads • Web serving • Containerized microservices • Media transcoding • Large-scale Java applications

Preguntas

- Cuales son las diferencias entre la serie N2 y la N4? Por qué elegimos una o la otra?
- Que significa que un tipo de máquina tenga en su nombre -lssd? Y que representan otras versiones de esta variación?
- Que significa que una serie termine en A o en D? Qué implica eso?



- GCP nos cobra por CPU si la máquina está en estado de RUNNING o PENDING_STOP
- GCP nos cobra por CPU si la máquina está en estado RUNNING, PENDING_STOP, SUSPENDING o SUSPEND
- Los attachable resources sobreviven a la máquina si pasa a estado TERMINATED y nos siguen cobrando por su uso (si corresponde)

Pregunta: Y los -lssd?

Provisioning Models

	Standard	Spot	Flex-start	Reservation-bound
Summary	• Based on resource availability, you can immediately create compute instances. • You can control when to stop or delete compute instances.	• Based on resource availability, you can immediately create compute instances. • You can control when to stop or delete compute instances. However, you also allow Compute Engine to stop or delete compute instances at any time to reclaim capacity.	• Based on resource availability, you can create compute instances within a specified waiting time. For a standalone Flex-start VM, you can specify a waiting time of up to two hours. For a MIG resize request, the waiting time is indefinite. • You can control when to stop or delete compute instances. However, you can't suspend or recreate them. Compute instances run for a minimum of 10 minutes and up to a maximum of seven days. When the compute instances reach the end of their run duration, Compute Engine stops or deletes them based on their termination action.	• You can request to reserve capacity at a future date for creating compute instances. If Google Cloud approves your request, then Compute Engine creates a reservation. At the start of the reservation period, you can consume the reservation by creating compute instances that match the reservation. • During the approved reservation period, you can stop, restart, delete, and recreate compute instances to consume the reservation as needed. When the reservation period ends, Compute Engine deletes the reservation, and stops or deletes any compute instances that consume the reservation based on their termination action.

Provisioning Models

Use cases	Ideal for workloads that require stability and continuous operation, such as the following workloads:
	• Web servers
	• Databases
	• Enterprise applications
	• Development and testing
		Ideal for workloads that can tolerate interruptions, such as the following workloads:
		• Batch processing
		• High performance computing (HPC)
		• Continuous integration and continuous deployment (CI/CD)
		• Data analytics
		• Media encoding
		• Online inference
			Workloads that require stability and need to run for no more than seven days, such as the following workloads:
			• Small model pre-training
			• Model fine-tuning
			• HPC simulation
			• Batch inference
				Ideal for workloads that require stability and a specific run time, such as the following:
				• **For workloads that last up to 90 days**:
				  • Model pre-training jobs
				  • Model fine-tuning jobs
				  • HPC simulation workloads
				  • Short-term expected increases in inference workloads
				• **For workloads longer than 90 days**:
				  • Training workloads
				  • Inference workloads

Provisioning Models

Supported machine series	You can use any machine series, except A4X Max, A4X, A4, and A3 Ultra.	You can use any machine series, except A4X instances and any bare metal instances (A4X Max, C4D, C4, C3, X4, and Z3).	You can only use the following machine series: • A4, A3, A2, G4, and G2 machine series • N1 virtual machine (VM) instances with GPUs attached • H4D machine series	Based on how you reserve capacity to create VMs, you can only use the following machine series: • If you reserve capacity in AI Hypercomputer, then you can only use A4X Max, A4X, A4, A3 Ultra, A3 Mega, A3 High with 8 GPUs, A3 Edge, and H4D machine series. • If you create a future reservation in calendar mode, then you can only use A4, A3 Ultra, A3 Mega, A3 High with 8 GPUs, or H4D machine series.

Compute Engine Calculator

MIGs

MIGs
(Managed Instance Groups)

Managed Instance Groups

- Son N máquinas virtuales administradas como un conjunto.
- Todas tienen que ser idénticas (como logramos esto?)
- **High Availability**: nos ofrece automatic repairs basados en la salud de la VM y en la salud de la aplicación. Además nos da soporte regional para el deployment de las VMs, de forma de que si falla una zona nuestro servicio sigue funcionando (esto es más caro?). Con esta estructura podemos usar un load balancer y distribuir la demanda.
- **Scalability**: ante cierto escenario el sistema puede levantar instancias nuevas para responder a picos de demanda
- **Automatic Updates**: podemos hacer rollouts de diferentes versiones de nuestra aplicación